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Modelling of Over-The-Rotor Acoustic Treatments for Improved Noise Suppression in Turbofan Engines

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Fan intake liner performance is likely to be reduced in the next generation of higher bypass ratio turbofan engines as a result of the expected decrease in the intake liner length (L) to fan diameter (D) ratio. It may be possible to recover engine noise levels by treating currently unlined surfaces in the immediate proximity of the fan rotor, in particular, with liners over the fan rotor tip. This Over-The-Rotor (OTR) liner concept has been studied recently, mainly at NASA GRC, and significant noise reductions have been measured. According to the literature, the noise attenuation mechanism caused by the OTR liners is attributed to a combination of source modification (by acting as a pressure-release surface), conventional absorption of acoustic waves propagating over the liner and a reduction in the rotor-stator interaction noise, although a full understanding of the physical mechanisms has yet to be achieved. This paper describes efforts at the ISVR to improve our understanding of OTR liner acoustic performance by using analytical models based on monopole and dipole Green's functions for lined cylindrical ducts. The models show that when sources are located at positions representative of a fan rotor tip source, an OTR liner can yield a peak power insertion loss of around 4 dB, with a significant broadband insertion loss over a wide range of source excitation frequencies. An impedance model of acoustically treated circumferential grooves coupled with a propagation analytical model predicts values of peak SPL acoustic attenuation in the forward arc of 3 dB. The analytical estimates present a good agreement with the W-8 NASA OTR data in the peak attenuation but with significant disagreement in the spectral shape.

I. Nomenclature

a	= duct radius
c_0	= speed of sound
d, h	= groove and cavity liner depth
f	= dipole source strength
G	= Green's function of the pressure
I, P	= acoustic intensity and power
J_m, Y_m	= Bessel function of the first and second kind of order m
m, n	= circumferential and radial modal orders
M	= Mach number
p, u	= acoustic pressure and particle velocity
Z	= specific acoustic impedance
x, r, θ	= axial, radial and circumferential coordinate
x_s, r_s, θ_s	= source axial, radial and circumferential coordinate
α, κ	= radial and axial wave number
γ	= dipole alignment w.r.t the duct axis
ω, He	= Helmholtz number based on the duct radius
ρ	= density

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II. Introduction

THE growth in air traffic increases the levels of noise in the population, which require the implementation of new technologies to accomplish the noise regulations in Europe, United States and worldwide. The aircraft noise generated during take-off and landing is key for the certification of aircraft. In modern turbofan engines, fan noise is one of the dominant sources both at approach and take-off. Hence, fan noise remains a critical noise-reduction target for the next generation of turbofan engines.

Fan intake liner performance will be reduced in the next generation of higher bypass ratio turbofan engines as a result of an expected decrease in the intake liner length (L) to fan diameter (D) ratio. On top of that, the potential use of composite materials for the future fan blade designs could lead to an increase of the tip clearance to avoid the extended blade to rub into the casing, which would increase the strength of the tip vortex and, therefore, the noise generated. In order to recover engine noise levels, every effort is needed to apply liners to other available surfaces. An area that has not been fully evaluated or exploited in the past is the duct casing in the immediate proximity of the fan rotor. The so-called fan proximity liners consisting of the fan case, the Over-The-Rotor (OTR) region and the interstage.

Among the Fan Proximity Liners, Over-The-Rotor acoustic treatments have been investigated during the last decade as a technology with the potential to further attenuate the fan noise in turbofan engines. These studies have been led by NASA in collaboration with other research centres, companies and universities. A summary of the noise attenuation performance and the impact in the fan aerodynamic efficiency is shown in Table 1. This paper is focused on the work in progress at the ISVR to improve the understanding of the acoustic attenuation of OTR liners through the development of analytical models and its validation with published experimental data.

Table 1 Summary of OTR published experimental results.

Publication	Noise Attenuation	Loss in Adiabatic Efficiency
Sutliff et. al. [1]	4 dB PWL (inlet & aft)	-
Elliot et. al. [2]	1 dB OAPWL	-
Sutliff et. al. [3][4]	5 dB inlet PWL / 2.5 dB OAPWL	1-2 %
Hughes and Gazzaniga [5]	-	6.5 - 9.3 %
Bozak et. al. [6]	-	0.75 %
Gazella et. al. [7]	1 dB / 3 dB (inlet/aft) PWL	-
Bozak and Dougherty [8]	2.5-3.5 dB (inlet) PWL	-

III. Background

A. Physical Mechanism

The idea of placing the liners over-the-rotor is aimed at reducing the noise generated in the region of the fan tip by modifying the interaction between the unsteady fan tip flow and the fan case. Primarily it is understood that the liner is placed in the fan case over the fan blade tip to absorb the pressure fluctuations by different mechanisms explained below.

Experimental tests have indicated that OTR liners mitigate the acoustic near-field, showing that noise attenuation of this acoustic treatment is caused by a combination of acoustic attenuation and source modification [1]. To confirm this conclusion, a series of tests were developed with different liner concepts installed in OTR configuration and in the inlet location to assess the relative importance of the physical attenuation mechanisms. When placed in the OTR location, the liners produced a higher noise reduction and thereby indicated that the attenuation mechanism is also attributed to source modification [4]. These last experiments also tested the liner designs in two hardware configurations: with and without static vanes, to investigate the impact of this concept in reducing the rotor-stator interaction noise. The results indicated that the attenuation mechanism is not only attributed to source modification but also to a reduction in the propagating noise from the rotor-stator interaction.

In summary, the noise attenuation mechanism caused by the OTR liners is attributed to a combination of three factors: modification of the source itself, conventional absorption of acoustic waves propagating over the liner and a reduction in the rotor-stator interaction noise. However, a consensus on the physical mechanism for the noise attenuation using OTR

acoustic treatments has not been reached. Further investigation into understanding of the physical mechanisms and predicting the noise suppression caused by the OTR liners is required [4][9].

B. Noise Attenuation Performance

Prototypes of foam metal liners for OTR acoustic treatment were tested in the Advanced Noise Control Fan (ANCF) test bed [3], which is a low-speed fan used for validation of noise reduction concepts at the Aero-Acoustic Propulsion Laboratory (AAPL) at NASA John H. Glenn Research Center (GRC). The foam metal liners were tested at different axial positions, including OTR. The best acoustic performance was measured when placed in the OTR position, resulting in up to 4 dB of broadband attenuation in both inlet and aft far fields.

To evaluate the effect of the OTR acoustic treatments in a higher bypass ratio (BPR) turbofan, two liner concepts were tested in the Pratt & Whitney Advanced Ducted Propulsor (ADP) as it is representative of an ultra-high bypass ratio, low tip speed turbofan engine [2]. One foam metal liner had the metal foam exposed on the flow surface and the other had a perforated rigid plastic resin rub surface; both were compared with a conventional rigid hard wall baseline. It was found that the OTR with a cover sheet showed an Overall Sound Pressure Level (OASPL) reduction of around 1 to 2 dB for this high bypass ratio hardware, which is lower than found in the previous experiments. However, the treatment with the exposed metal foam produced more noise than the hard wall baseline case, around 1 dB higher.

A Williams International FJ44-3A was used as a test bed at NASA Glenn to assess the effect on the performance of foam metal liners located OTR in a high-speed turbofan engine and its attenuation benefits were compared against a hard wall baseline, using typical SDOF liners designed using standard methodologies [3, 4]. The foam metal liner was less effective at and above sonic fan tip speed, which may be due to the shock waves generated at the fan blades at supersonic tip speeds. However, the SDOF liners located at the traditional inlet position maintained its effectiveness above sonic tip speed. It should be noted though, that the foam metal liner acoustic attenuation relative to the liner length is generally higher. Up to 5 dB attenuation in *inlet* Power Level (PWL) was achieved below fan tip sonic speed, with a total OAPWL reduction of 2.5 dB. The use of circumferential grooves in the OTR liners showed a smaller performance penalty, which is explained in the following pages. However, it was found that grooves generate additional broadband noise that is not fully attenuated by the OTR liner [9].

The reduction in aerodynamic performance penalty observed in previous publications [6] was attributed to the circumferential grooves in the casing installed over the fan tip and its effect on reducing the dynamic pressure on the acoustic treatment. This lead to liner designs consisting of a casing of circumferential grooves integrated with various acoustic treatments [8]. Tests were developed aiming to improve the understanding of the physical mechanisms of noise attenuation using OTR acoustic treatments to improve the Technology Readiness Level (TRL) of various OTR liner designs in the ANCF [7] and in the W-8 Single Stage Axial Compressor Facility [8], both installations at NASA GRC.

In the ANCF tests, the liner designs were (1) compared against a hard-wall baseline, (2) installed in OTR configuration and in the inlet to assess the relative importance of the physical attenuation mechanisms and (3) tested with and without static vanes to investigate the impact of this concept in reducing the rotor-stator interaction noise. The best liner concept achieved the highest reductions, up to 3dB, in the mid-frequency range (3-4th harmonic bands). The thin face sheet was a common factor in the designs with higher attenuation. Some treatments showed an increase in noise when comparing the results of rotor-alone vs rotor-stator, which was attributed to an increase in strength of the rotor tip vortex impinging on the stator vanes.

The results from the W-8 Single Stage Axial Compressor showed a net benefit of the OTR liner designs of 2.5-3.5 dB of reduction in the PWL of the forward propagating modes. A noise penalty of 2.5-6 dB from the circumferential grooves was present between 4-8 kHz at low fan speeds. The impact of the grooves at high fan speeds was found to be minimal [8]. Recent Grazing Flow Impedance Tube tests at NASA showed that the additional groove noise is likely cavity tones/resonances that are clipped by the fan *.

C. Aerodynamic Performance

The aim of reducing the noise generated by the engine needs to be traded off against any loss of performance. A number of investigations have been developed to assess the effect of the OTR liner concepts on the aerodynamic performance of the engine and to optimize the design for a minimal aerodynamic penalty while presenting the desired noise attenuation [4–6].

*Results to be published at the 25th AIAA/CEAS Aeroacoustics Conference (2019) by Bozak et. al. under the title "Effect of Grazing Flow on Grooved Over-the-Rotor Acoustic Casing Treatments".

In one of the investigations [5] two OTR metal foam liners were tested, one that had the metal foam exposed on the flow surface and the other with a perforated rigid plastic resin rub. Both were compared to a conventional rigid hard wall baseline. The OTR perforated rubstrip showed a drop in performance with respect to the baseline case with a loss in adiabatic efficiency from 2.7% to 4.0% as the speed increases. The exposed metal foam OTR showed even worse performance, with a loss in adiabatic efficiency from 6.8% at 55% speed to 9.3% at 100% speed. Better results were obtained in the tests of FML OTR on the Williams International FJ44-3A [4], which showed a reduction of 1-2% in the engine performance, a significant improvement over previous tests.

Further work was undertaken to quantify the impact of OTR liners with circumferential grooves with respect to a baseline hard wall rub-strip in the aerodynamic performance of a turbofan rotor [6]. The variation in pressure ratio, fan temperature ratio and fan adiabatic efficiency were used to quantify the performance loss. The fan pressure ratio was not affected by the OTR treatment. However, the fan temperature ratio near the blade tips increased, with a rapid rise near sonic tip speeds. The adiabatic efficiency loss was 0.75%, indicating a benefit from using the circumferential grooves in the OTR liner designs to minimize the loss of aerodynamic performance.

IV. Liner Propagation Model #1

A. Formulation

The model consists of an infinite hollow circular duct which is excited by a point source. A non-zero uniform mean flow exists in the duct which has hard or lined wall boundaries. A sketch of the geometry is shown in Fig.1. The development of this model has set a foundation for liner propagation Model #2. The formulation and nomenclature are based on the analytic Green's function for a lined circular duct containing uniform mean flow published by Rienstra and Tester [10] and expressed as a sum of non-orthogonal modes.

In all analytical expressions of this paper, the axial distance is normalised by a , the frequency by c_0/a , the pressure by $\rho_0 c_0^2$, the particle velocity by c_0 and therefore the intensity is normalised by $\rho_0 c_0^3$ and the power by $\rho_0 c_0^3 a^2$.

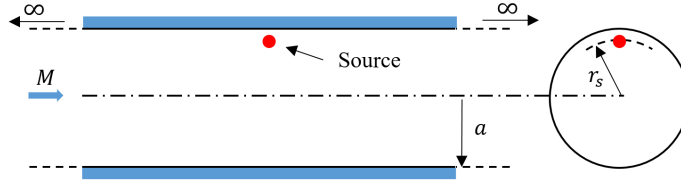


Fig. 1 Sketch of the lateral and cross-section of the infinite circular duct problem.

1. Static point monopole

The pressure field generated at $\mathbf{x} = (x, r, \theta)$ by a monopole source located at $\mathbf{y} = (x_s, r_s, \theta_s)$ in a lined circular duct with mean flow can be obtained by using the analytic Green's function [10]:

$$G(\mathbf{x}|\mathbf{y}) = \sum_{m=-\infty}^{\infty} e^{-jm(\theta-\theta_s)} G_m(r, x) \quad (1)$$

where

$$G_m(r, x) = -\frac{1}{2\pi j} \sum_{n=1}^{\infty} \frac{J_m(\alpha_{mn}^{\pm} r) J_m(\alpha_{mn}^{\pm} r_s)}{Q_{mn}^{\pm} J_m(\alpha_{mn}^{\pm})^2} e^{-j\kappa_{mn}^{\pm}(x-x_s)} \quad (2)$$

with

$$Q_{mn}^{\pm} = \pm \left[(\kappa_{mn}^{\pm} + \Omega_{mn}^{\pm} M) \left(1 - \frac{m^2}{\alpha_{mn}^{\pm 2}} - \frac{\Omega_{mn}^{\pm 4}}{(\omega \alpha_{mn}^{\pm} Z)^2} \right) - \frac{2jM\Omega_{mn}^{\pm}}{\omega Z} \right] \quad (3)$$

where

$$\Omega_{mn}^{\pm} = \omega - \kappa_{mn}^{\pm} M \quad (4)$$

$Q_{mn}^{\pm}$, $\alpha_{mn}^{\pm}$ and $\kappa_{mn}^{\pm}$ are used for the right-running (+) and left-running modes (-), in the regions $x > x_S$ and $x < x_S$ respectively. The axial and radial modal wave numbers can be obtained by finding the roots of:

$$j\Omega_{mn}^{\pm 2} J_m(\alpha_{mn}^{\pm}) + \omega \alpha_{mn}^{\pm} Z J'_m(\alpha_{mn}^{\pm}) = 0 \quad (5)$$

given that:

$$\alpha^2 = \Omega^2 - \kappa^2 \quad (6)$$

The expressions above are simpler for the hard wall boundary condition, for which Eq. 5 simplifies into $J'_m(\alpha_{mn}^H) = 0$ and the axial eigenvalues can be found by using:

$$\kappa_{mn}^{\pm} = \frac{-\omega M \pm \sqrt{\omega^2 - (1 - M^2)\alpha_{mn}^{H^2}}}{1 - M^2} \quad (7)$$

However, it is not trivial to solve the eigenvalue equation for the lined case (Eq. 5). A script of Rienstra has been used in the current instance. An explanation of the eigenvalue problem and the approach for solving it can be found in [11]. More details of the formulations and the analogous expressions for the annular section can be found in the literature [10].

The axial particle velocity, u_x , can be obtained from the modal expansion of the pressure field and the non-dimensional linearised momentum equation in the axial direction:

$$j\omega u_x + M \frac{\partial u_x}{\partial x} + \frac{\partial p}{\partial x} = 0 \quad (8)$$

which gives,

$$u_x(\mathbf{x}|\mathbf{y}) = \sum_{m=-\infty}^{\infty} e^{-jm(\theta-\theta_s)} \left(-\frac{1}{2\pi j} \sum_{n=1}^{\infty} \frac{\kappa_{mn}^{\pm}}{\Omega^{\pm}} \frac{J_m(\alpha_{mn}^{\pm} r) J_m(\alpha_{mn}^{\pm} r_s)}{Q_{mn}^{\pm} J_m(\alpha_{mn}^{\pm})^2} e^{-j\kappa_{mn}^{\pm}(x-x_s)} \right) \quad (9)$$

2. Static point dipole

A dipole source orientated a certain angle γ relative to the duct axis has been implemented to account for the lift fluctuations due to the turbulence impinging on the fan blade and the trailing edge noise, see Fig.2.

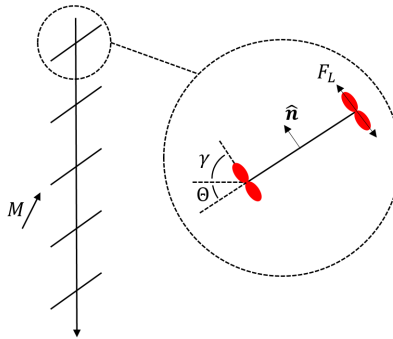


Fig. 2 Row of fan blades as viewed over the tip and the dipole source model, with γ the angle relative to the duct axis, Θ the stagger angle and F_L the fluctuating lift force.

The pressure field generated at $\mathbf{x} = (x, r, \theta)$ by a single dipole source of strength f located at $\mathbf{y} = (x_s, r_s, \theta_s)$ in a lined circular duct with mean flow can be obtained by using the analytic Green's function solution of the wave equation:

$$p(\mathbf{x}|\mathbf{y}) = f \hat{\mathbf{n}} \cdot \nabla G(\mathbf{x}|\mathbf{y}) \quad (10)$$

where G is defined by Eq. 1 and the dipole alignment $\hat{\mathbf{n}}$ normal to the blade surface S is given by

$$\hat{\mathbf{n}} = [\hat{x}, \hat{r}, \hat{\theta}] = [\cos \gamma, 0, \sin \gamma] \quad (11)$$

The nabla operator in cylindrical coordinates applied to the Green's function 1 gives:

$$\nabla G = \frac{\partial G}{\partial x_s} \hat{x}_s + \frac{\partial G}{\partial r_s} \hat{r}_s + \frac{1}{r_s} \frac{\partial G}{\partial \theta_s} \hat{\theta}_s \quad (12)$$

Assuming $f = 1$, the pressure and particle velocity result in

$$p(\mathbf{x}|\mathbf{y}) = \sum_{m=-\infty}^{\infty} e^{-jm(\theta-\theta_s)} \left(-\frac{1}{2\pi} \sum_{n=1}^{\infty} \frac{J_m(\alpha_{mn}^{\pm} r) J_m(\alpha_{mn}^{\pm} r_s)}{Q_{mn}^{\pm} J_m(\alpha_{mn}^{\pm})^2} e^{-j\kappa_{mn}^{\pm}(x-x_s)} \left[\kappa_{mn}^{\pm} \cos \gamma + \frac{m}{r_s} \sin \gamma \right] \right) \quad (13)$$

and

$$u_x(\mathbf{x}|\mathbf{y}) = \sum_{m=-\infty}^{\infty} e^{-jm(\theta-\theta_s)} \left(-\frac{1}{2\pi} \sum_{n=1}^{\infty} \frac{\kappa_{mn}^{\pm}}{\Omega_{mn}^{\pm}} \frac{J_m(\alpha_{mn}^{\pm} r) J_m(\alpha_{mn}^{\pm} r_s)}{Q_{mn}^{\pm} J_m(\alpha_{mn}^{\pm})^2} e^{-j\kappa_{mn}^{\pm}(x-x_s)} \left[\kappa_{mn}^{\pm} \cos \gamma + \frac{m}{r_s} \sin \gamma \right] \right) \quad (14)$$

where $Q_{mn}^{\pm}$ and $\Omega_{mn}^{\pm}$ are the same than for the monopole Green's function. The sign logic and nomenclature also applies here. An equivalent solution including a general distribution of rotating dipoles is given in [12].

3. In-duct sound power

The non-dimensional acoustic axial power at each axial cross-section (S) can be used to assess the performance of liners. It is computed by integrating the intensity field over the duct cross-section, resulting in the expression [13–15]:

$$P(x) = \int_S I_x dS = \frac{1}{2} \int_S \text{Re} \left\{ p u_x^* (1 + M^2) + M p p^* + M u_x u_x^* \right\} dS \quad (15)$$

It is common to analyse the liner acoustic performance not directly with the power but using the Sound Power Level (PWL) Insertion Loss (IL) or just IL, defined as the power relative to the hardwall case in dB scale.

$$IL(x) = 10 \log_{10} \frac{P_H(x)}{P_L(x)} \quad [\text{dB}] \quad (16)$$

where P_H and P_L are the acoustic power for the hard wall and lined configurations obtained using Eq.15.

As usual, the infinite number of modes that define the analytical solution of the acoustic field cannot be computed. Therefore, a parameter to control the series truncation is used, which determines the number of modes considered in the computation. This parameter is called Cut-Off Ratio (COR) and the definition that has been used is:

$$COR = \frac{\alpha \sqrt{1 - M^2}}{\omega} \quad (17)$$

The radial wave number increases with the mode order, which means that for a given COR all the modes with α smaller than $\frac{\omega COR}{\sqrt{1 - M^2}}$ are included in the modal summation. Note that $COR = 1$ corresponds to only cut-on modes, and that $COR > 1$ will include a certain number of evanescent modes. The evanescent modes play an important role in the near-field of the source and therefore CORs sensibly higher than one have been used in the OTR model studies.

B. Effect of the boundary conditions on the source power output

In this section, an analysis is presented of the effects that the impedance boundary condition and the source radial position and excitation frequency have on the acoustic power generated by a monopole point source. The axial acoustic power is computed at the *source plane* by integrating the axial intensity over the cross-section of the duct. The intensity field is obtained from the Green's function modal expansion of the pressure and the axial particle velocity. This is enough for the hard wall case, as no energy is dissipated at the boundaries, and for the lined case, as the power is evaluated at the source plane omitting any energy dissipated by the liners. The axial power results of this section are normalised by the power radiated by a monopole source in the free-field. That is, P_d/P_f , the subscripts d and f referring to ducted and free-field respectively. An extensive analysis including correlated and uncorrelated distributions of monopoles and axial dipoles has been previously published for a hard wall case [16].

The effect of the source radial position and the excitation frequency in the normalised power output is shown in Fig.3 considering a hard wall case. The results indicate that the power tends to the free-field value when the source is located away from the wall and tends to twice this value when located at the wall itself. It can be seen that the wall acts as an image source if the source is located very close to the wall. The variation in radiated power gets progressively confined at the vicinity of the wall as the frequency increases. However, the effects of the source position are minimal in the low frequency range, where the power spectra is dominated by the resonances at the cut-on frequencies. This behaviour occurs both for a monopole and a dipole source, the latter not covered in this instance for brevity.

The analysis is extended to lined ducts, for which three values of impedance have been used to see the impact of this parameter on the trends observed in Fig.3. The effect of the lined surfaces is shown in Fig.4. The trends are similar to those for the hard wall case: the duct wall has almost no effect in the power output provided that the distance between the source and the wall is big enough, this distance depending on the source excitation frequency. This result is true for all values of impedance tested and also applicable to a dipole source. However, if the source is located close enough to the wall or the frequency is low enough, the value of the impedance has a high impact on the power output.

The results in Fig.3a and Fig.4a are plotted in Fig.5 in terms of the radial distance from the source to the wall normalised by the wavelength (λ). The curves collapse and show the relation between the source radial location and frequency outlined above. The source proximity to the wall is only important if the source is located within 0-0.3 wavelengths from it. In those cases, the impedance of the wall has a significant impact on the source power output.

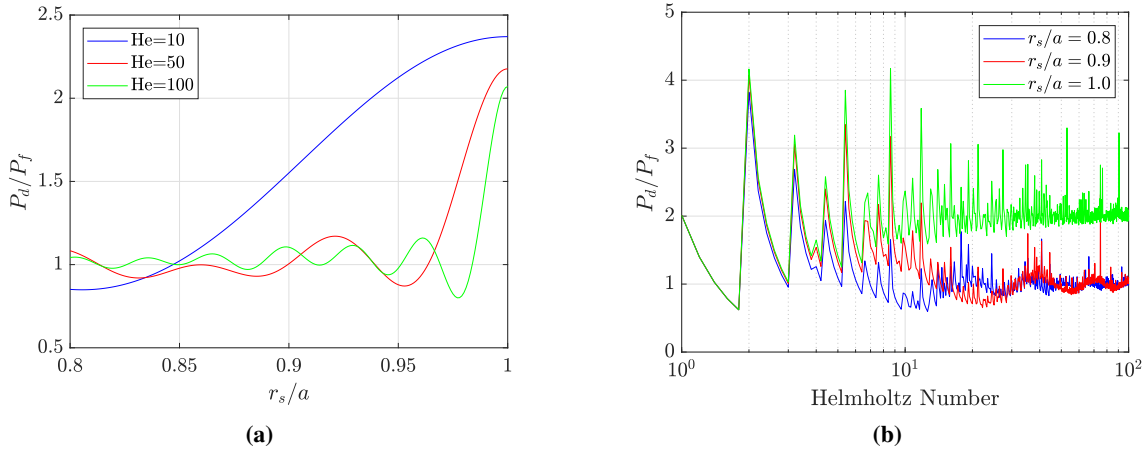


Fig. 3 Power generated by a monopole in a hard duct for various radial source positions and excitation frequencies with $COR = 5$ and $M = 0$.

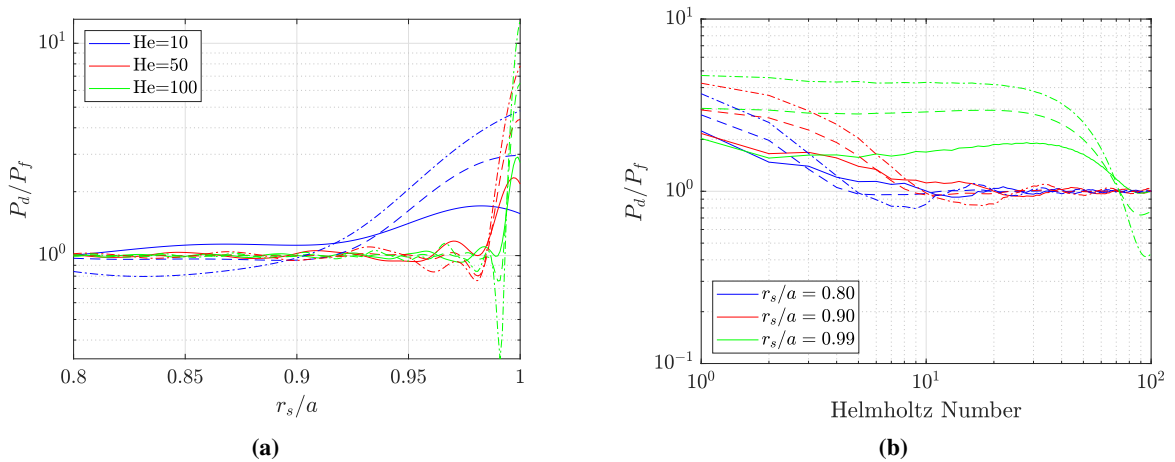


Fig. 4 Power generated by a monopole in a lined duct for various radial source positions and excitation frequencies with $COR = 5$ and $M = 0$. Solid line: $Z = 1 + j$, dashed line: $Z = 1$ and dash-dotted line: $Z = 1 - j$

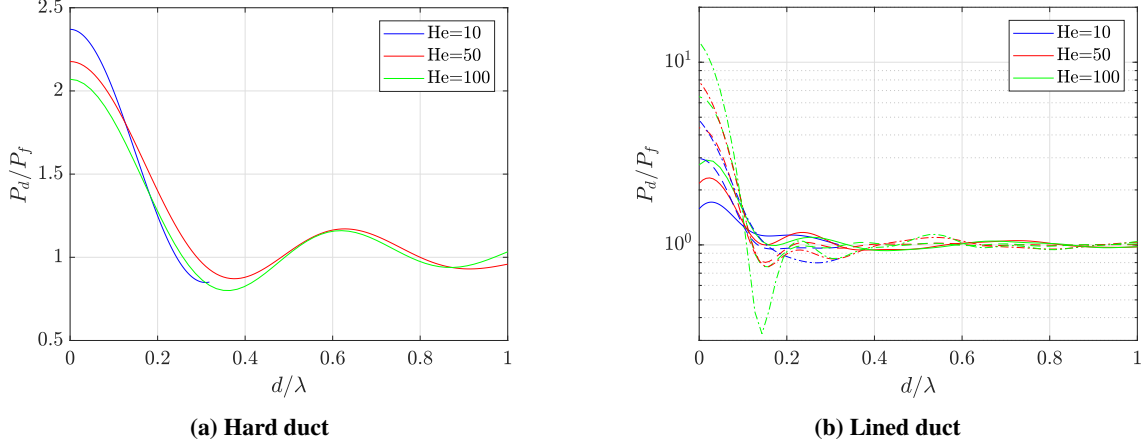


Fig. 5 Power generated by a monopole in a hard/lined duct for various normalised radial source positions ($d = 1 - r_s$) and excitation frequencies with $COR = 5$ and $M = 0$. Key of Fig.(b): solid line: $Z = 1 + j$, dashed line: $Z = 1$ and dash-dotted line: $Z = 1 - j$.

The effect of the liner in the power generated by the source can result in cases in which negative IL occur (amplification). That is, the variation in power generated by the source is higher than the attenuation of the relatively short liner. Therefore, a given power IL can be separated into two components: the contribution of the liner attenuation and the contribution of the source modification.

$$IL(x) = IL_{source} + IL_{liner}(x) = 10 \log_{10} \frac{P_H}{P_L(x_s)} + 10 \log_{10} \frac{P_L(x_s)}{P_L(x)} \quad [dB] \quad (18)$$

To show the aforementioned effect, these two contributions and the total PWL IL are computed for a range of liner resistances and reactances and are shown in Fig.6. As a sanity check, it can be seen that the attenuation of the liners always provides a positive power IL. For this particular case, Fig.6b shows that the range of power IL achieved by the liner attenuation is from 0.4 dB up to 3.3 dB and Fig.6c indicates that the impedance can modify the power generated by the source such that the power IL ranges from -1 dB to 0.8 dB. By adding the two contributions one would get the total power IL map shown in Fig.6a.

This model has shown that the power generated by a monopole source in a duct depends on the impedance boundary condition and the source excitation frequency and radial position. Now, the question is whether this behaviour also occurs in the case of a real fan rotor source.

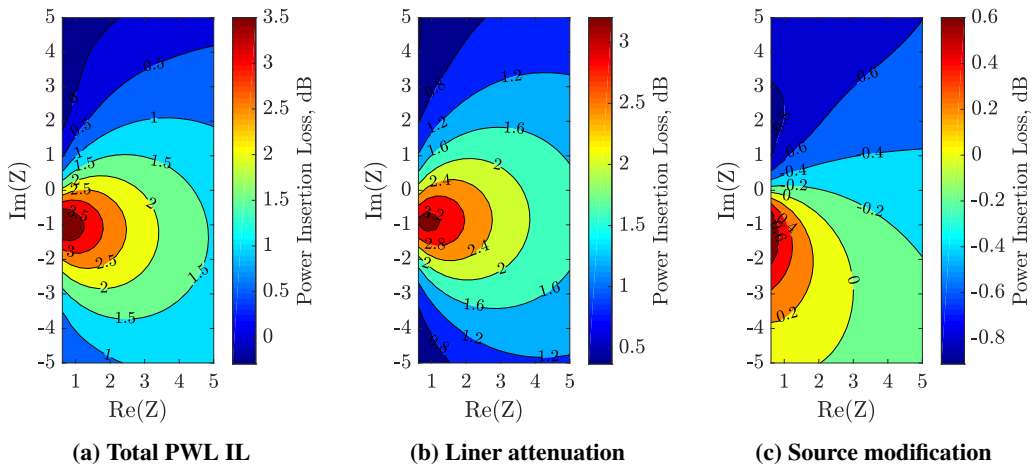


Fig. 6 Contour maps of the total PWL IL and the contributions of the liner attenuation and source modification, evaluated at $x = x_s - 0.068a$ using Green/INF with $r_s/a = 0.95$, $He = 31.35$, $COR = 5$ and $M = 0$.

V. Liner propagation Model #2

A. Formulation

The problem considered consists of a cylindrical semi-infinite lined duct matched with a semi-infinite hard wall duct, the pressure field being generated by a monopole/dipole point source. A diagram of the problem is shown in Fig.7. The pressure field generated by the source can be obtained by using the previous model (Model #1) and solving the matching problem at the interface between the lined and hard regions. By using this approach, one can modify the radial and axial position of the source to define the tip clearance and lined surface available respectively and then compute the axial power radiated into the upstream hard wall section to assess the performance of the liner. Furthermore, it allows studying any 'back-reaction' effects on the source that the proximity of the matching interface may cause.

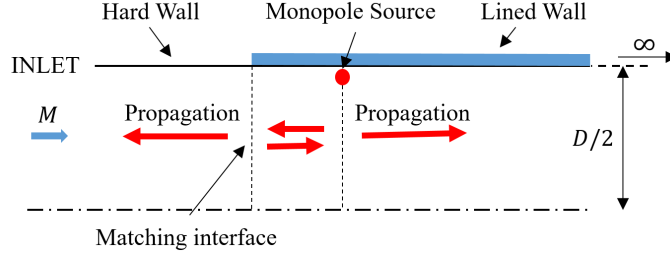


Fig. 7 Diagram of the semi-infinite hard-lined duct.

The mode-matching procedure is based on the continuity of mass and momentum at the matching plane or interface. This approach was proposed by Gabard and Astley [17] and it improves the accuracy of the mode-matching scheme in the presence of flow, especially at the vicinity of the impedance discontinuity. In the absence of flow, this mode-matching technique is equivalent to the standard method of continuity of pressure and axial particle velocity. Similar standard matching methods are described in the literature [18][19], including a comprehensive formulation of McAlpine et. al. [20]. The incident modes generated by the source are known from the Green's function and the transmitted and reflected modes are obtained by solving the matching equations. Modes coming from the upstream and downstream ends are neglected by imposing an anechoic condition at both ends. Details of the matching formulation are given in Appendix A.

B. OTR Performance Estimates

The matching plane is located at $x = 0$ and the axial position of the source is fixed at $x_s/a = 0.068$. This axial distance is half the length of the space available over the rotor in the W-8 test rig. That is, it assumes that the monopole/dipole source is located at mid-chord of the fan blade

It has to be noted that, as the source moves towards the matching plane, a higher number of evanescent modes are required to have an acceptable convergence of the solution (higher COR required). In addition, as the modes attenuation depends on the impedance, different wall impedances require different COR to converge. Therefore, to ensure that the solutions are reliable, a study of the COR necessary to converge for a fixed source excitation frequency was developed. The results for a monopole source can be observed in Fig.8a, which show that a higher COR is required for low resistances and reactances. For the cases of interest in this instance, where $x_s/a = 0.068$, it has been found that a $COR = 5$ is generally enough to provide converged results for the relevant ranges of radial positions ($0.8 \leq r_s \leq 1$) and source excitation frequencies ($1 \leq He \leq 50$). This statement is also extensible to cases with a dipole source. If a distribution of axial sources along the fan chord or other smaller axial source positions were to be tested, the COR should be corrected accordingly to obtain accurate results in the mode-matching computation. Further work will include an analysis of the required COR for convergence as a function of x_s and a given convergence condition.

PWL Insertion Loss (IL) estimates have been computed for a range of impedances assuming a monopole and a dipole source, the latter with $\gamma = 30$ deg. The resulting converged IL impedance map for $He = 31.35$, $x_s/a = 0.068$, $r_s/a = 0.95$ and $M = 0$ is shown in Fig.8b and Fig.8c. The monopole results indicate a "near-optimum" impedance of $Z = 0.62 - 1.13j$ that provides 5.5 dB of power insertion loss for the input conditions used. The dipole "near-optimum" impedance for this conditions is $Z = 0.62 - 0.62j$, which provides 5 dB of power IL. Different excitation frequencies or radial source positions will modify the source power output, as explained in Section IV, and therefore will change this attenuation contour plots.

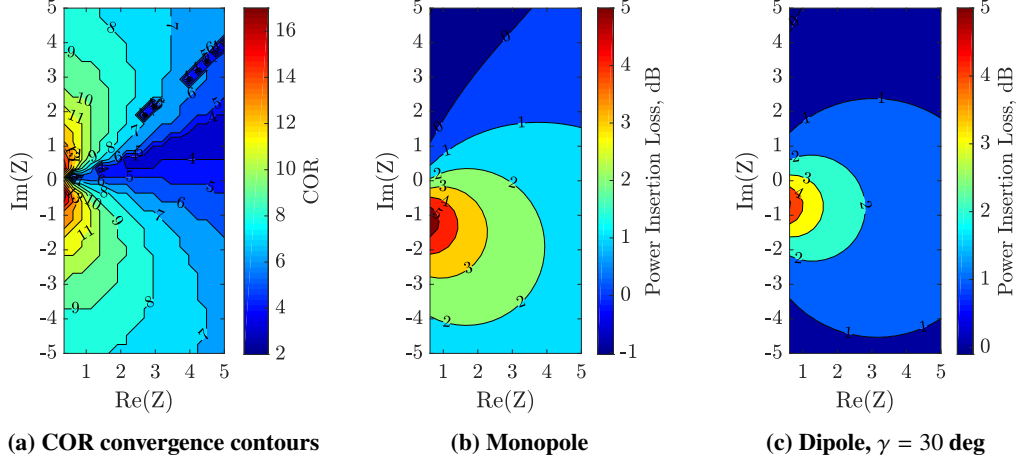


Fig. 8 (a) COR required to convergence and (b)-(c) power IL impedance contour plots for $COR = 5$, $x_s/a = 0.068$, $r_s/a = 0.95$, $He = 31.35$ and $M = 0$ for a monopole and dipole source.

An important consideration in the feasibility of the OTR concept is to make sure that it can operate over a wide range of frequencies without having a negative impact on the overall noise attenuation performance. To do so, a simple study has been developed looking at how the power IL changes with different excitation frequencies considering a SDOF locally reacting liner, modelled by:

$$Z(\omega) = R_f + j \left(k \frac{t+d}{\sigma} - \cot(kh) \right) \quad (19)$$

where k is the wave number, R_f the non-dimensional facing sheet resistance, t and d the facing sheet thickness and hole diameter respectively, h the liner depth and σ the facing sheet porosity. The particular values used are shown in Table 2.

Table 2 Numerical values used for the SDOF cavity liner.

R_f	t	d	σ	h
1.5	0.8 mm	1 mm	0.05	25 mm

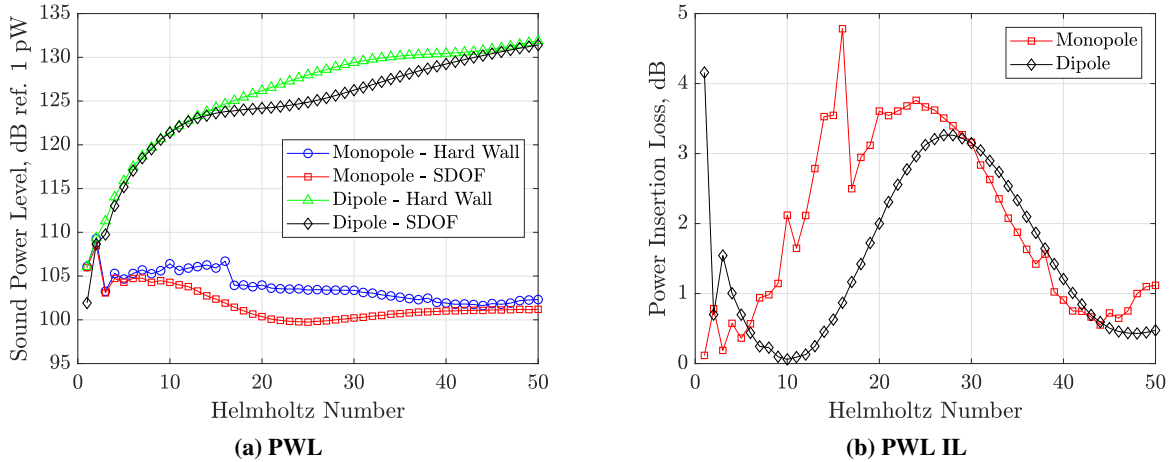


Fig. 9 Variation of (a) the PWL and (b) power IL with the source excitation frequency for a monopole and an axial dipole located at $x_s/a = 0.068$ and $r_s/a = 0.95$ with $M = 0$ and $COR = 5$ assuming a SDOF cavity liner.

After an appropriate convergence check, the results obtained are shown in Fig.9 for a monopole and an axial dipole. In the monopole case, these estimates show a peak power insertion loss (IL) of around 4 dB, with significant attenuation over a wide range of source excitation frequencies. It should be noted that the noise generated is not increased by using this SDOF OTR configuration. When an axial dipole is used, the peak attenuation is approximately 1 dB lower but still effective over a quite wide range of frequencies. The reduced attenuation of the dipole can be attributed to the directivity of this source model.

An increase in the space available from the source to the matching plane has as a consequence an increase in the IL, as the lined area available is higher, as one would expect. However, the reflected modes from the matching interface can have a 'back-reaction' effect on the source, thus modifying its radiation efficiency. To assess this effect, one can subtract the power IL contour maps evaluated at $x = x_s$ using Model #1 and those obtained with Model #2 considering a distance x_s between the source and the matching plane. Since Model #1 does not have any matching interface, any difference should be caused by the impedance discontinuity.

Fig.10 show the difference $IL_{M2} - IL_{M1}$ when the source is located close to the matching plane (a) and one radius away from it (b), the subscripts referring to Method #2 and Method #1 respectively. When the source is located far from the interface the variations between the two methods are minimal, whereas they can vary significantly (-0.6 to 1.4 dB) when it is located close to it. This result highlights the 'back-reaction' effect of the reflected modes on the source previously mentioned.

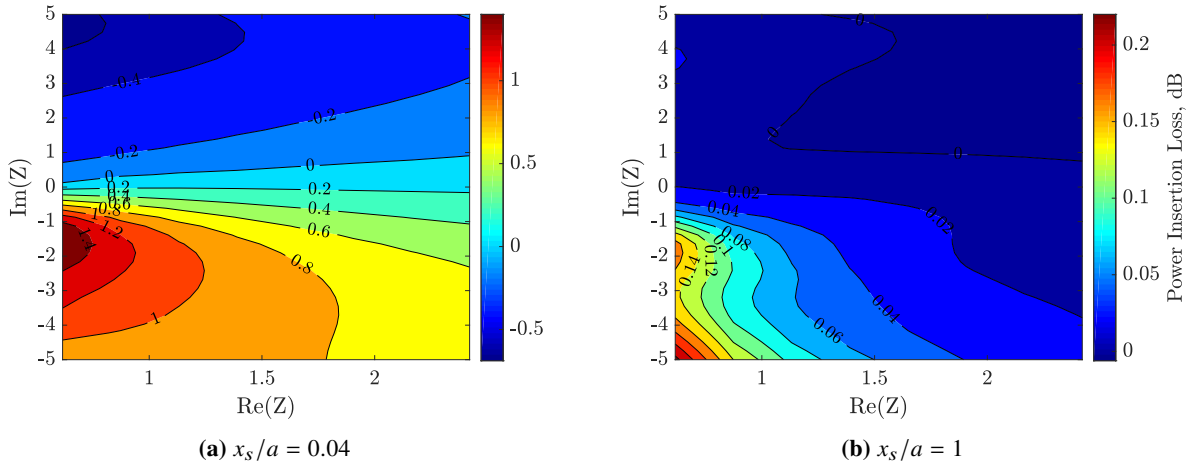


Fig. 10 Difference of the power IL impedance maps obtained using Method #1 and Method #2 for a monopole source located at $x_s/a = 0.04$ and $r_s/a = 0.95$ with $He = 31.35$, $COR = 5$ and $M = 0$.

VI. Modelling of acoustically treated circumferential grooves

The analytical OTR prediction models presented in this paper are compared, in Section VII, with the experimental data of the latest OTR circumferential groove array tested in the NASA W-8 fan rig, published by Bozak and Dougherty [8]. Therefore, an equivalent impedance boundary condition that meets that of the published experimental data has to be used in the analytical predictions. These analytical models and their cross-verification using FEM simulations are outlined in this section. The modelled OTR fan case configurations consist of circumferential grooves that cover the axial projection of the fan chord, with the bottom of the grooves treated with cavity liners or hardwall respectively. Fig.11a shows a diagram of the OTR circumferential grooves with a cross-section of the acoustic treatment.

The analytical impedance model presented in this instance currently neglects any mean axial or azimuthal flow in the groove. The groove is assumed to be non-locally reacting in the azimuthal direction and locally reacting in the axial direction. For the case of acoustically treated grooves, the impedance at the septum is treated as a locally reacting SDOF cavity liner. The detailed assumptions and the formulation of the model can be found in Appendix B. The effective impedance of the treated grooves resulting from this model is

$$Z_g(\omega, m) = -j \frac{J_m(\omega) + K_m Y_m(\omega)}{J'_m(\omega) + K_m Y'_m(\omega)} \quad (20)$$

where

$$K_m = -\frac{J'_m(\omega[1+d]) + j\frac{J_m(\omega[1+d])}{Z_f}}{Y'_m(\omega[1+d]) + j\frac{Y_m(\omega[1+d])}{Z_f}} \quad (21)$$

with Z_f being the effective impedance at the septum. That is, a SDOF cavity liner response in the lined case or infinity for the hard groove configuration.

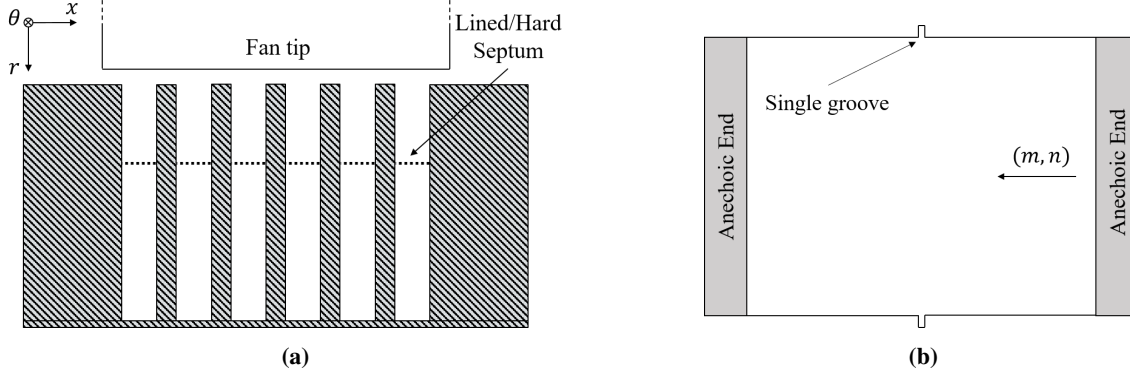


Fig. 11 (a) OTR circumferential groove array as tested in NASA W-8 fan rig and (b) Virtual.Lab model used in the simulation of a single groove problem.

A numerical case study using the FEM software LMS Virtual.Lab has been performed to assess the acoustic response of a simplified groove geometry to various incident duct modes. The ultimate aim is to verify that the assumptions made in the groove analytical model and the model itself are valid for cases of interest. The numerical case consists of an infinite cylindrical duct with a single groove and excited by incident duct modes of azimuthal mode order m and radial mode order n . A diagram of this case is depicted in Fig.11b. The infinite duct has been numerically modelled by using an anechoic boundary condition at each end of the duct section. The geometry of the groove is defined from a CAD model and set to meet the dimensions of the grooves tested in the NASA W-8 fan rig. A boundary condition of SDOF locally reacting liner is imposed at the bottom of the groove

A complete analysis of numerical results is not presented in this paper and will be covered in future publications. However, it has been observed that the axial variation of the acoustic variables within the groove is minimal, showing a plane-wave behaviour and justifying the assumption of axially locally reacting. Therefore, these can be averaged in the axial direction to give a numerical solution of the form $p(\omega, m, n, r)$ & $u_r(\omega, m, n, r)$, which can be used to obtain a numerical effective acoustic impedance $Z_{num}(\omega, m, n, r)$ directly comparable to the analytical model. The equation of the groove effective impedance derived from the semi-locally reacting analytical model is based on a continuous solution of the wave equation throughout the groove and therefore the radial impedance at any point *within* the groove can be computed by evaluating the pressure and radial particle velocity at any radius:

$$Z_g(\omega, m, r) = -j\frac{J_m(\omega r) + K_m Y_m(\omega r)}{J'_m(\omega r) + K_m Y'_m(\omega r)} \quad (22)$$

A comparison of the analytical and numerical effective impedance and admittance ($A = 1/Z$) is shown in Fig.12 and Fig.13 for a hard and lined duct respectively, limited to modes (5,1), (15,1) and (20,1) for clarity. In the hard groove case, both the resistance and the reactance results show a good agreement, tending to minus infinity when moving towards the hard wall boundary condition. The agreement is clearer in the admittance plot, where it can be seen that both the real and imaginary part tend to zero at the hard wall as expected.

In the lined groove case, both the resistance and the reactance satisfy the imposed boundary condition at the septum and generally present a satisfactory agreement, especially from half the groove depth to the groove end. The disagreements between the analytical and numerical impedance are usually higher close to the interface main duct-groove or groove 'neck', with significant variations of up to $1.5\rho_0 c_0$. A refinement of the mesh in the region of the 'neck' is planned in order to improve the resolution and accuracy of the simulation and to then reassess the present analysis.

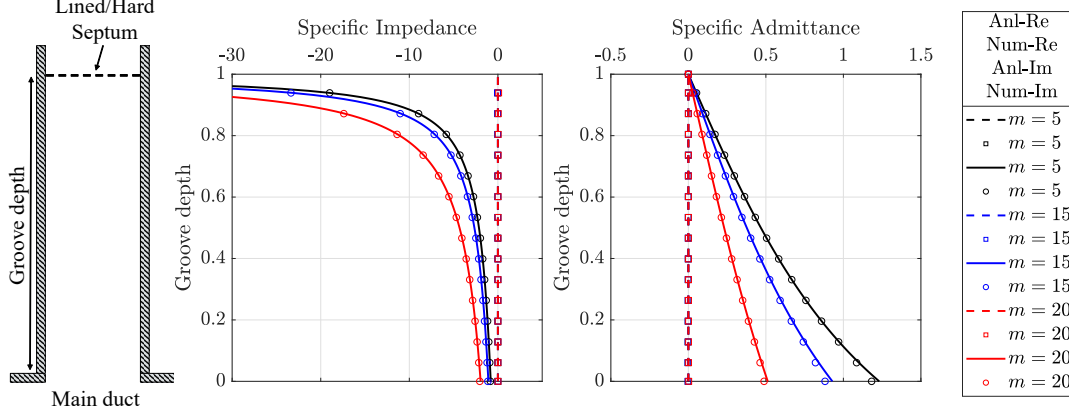


Fig. 12 Radial variation of radial impedance and admittance within groove for different incident azimuthal modes: comparison of analytical model and numerical results for hard groove at 3900 Hz.

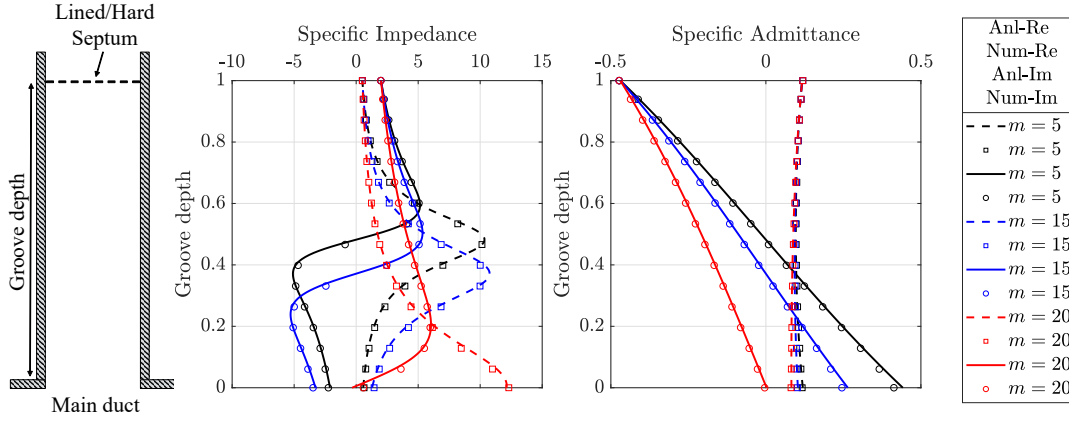


Fig. 13 Radial variation of radial impedance and admittance within groove for different incident azimuthal modes: comparison of analytical model and numerical results for lined groove at 3900 Hz.

VII. Comparison of Model #2 with NASA experimental data

In this section, the results obtained with Model #2 coupled with our analytical groove impedance model are compared with the latest OTR experimental data published by Bozak and Dougherty [8]. The analytical calculation presented in this section considers an axial dipole static point source. The circumferentially averaged sound pressure level (SPL) at the duct wall has been used to compare the analytical estimates with the experimental data. The SPL is evaluated at $2.2a$ upstream of the end of the OTR acoustically treated casing, the position where the circumferential microphone array is installed in the test rig. This magnitude gives an indication of the acoustic in-duct PWL.

The modelling of the equivalent impedance of configurations 'empty chamber' and 'groove-only baseline' is outlined in Section VI and detailed in Appendix B. For the sake of clarity, the experimental datasets are re-named as 'lined grooves', 'hard grooves' and 'hard wall' respectively. The coupling of Model #2 with our analytical impedance model cause problems in the eigenvalue solver for certain cases where the resistance is very small. This is especially problematic for the 'hard grooves' configuration since it is modelled as a purely reactive impedance. An 'artificial' resistance of $R_a = 0.1$ has been added to the effective impedance of both grooved configurations to mitigate this issue. This resistance can be linked to viscous dissipation at the groove neck and has a small effect (< 0.5 dB) on the SPL IL predictions. An additional correction based on the continuity of mass in the radial direction has been applied to the effective impedance to account for the solid surfaces between the grooves cavities. These two corrections can be expressed as

$$Z_{eff} = R_a + \frac{S_t}{S_g} Z \quad (23)$$

where Z is calculated as per Eq.20 and S_t , S_g are the total surface of the acoustic treatment and the total surface of interfaces groove-duct respectively. The effective impedance of the hard and lined grooves configurations predicted with this analytical groove impedance model is shown in Fig.14 for different azimuthal mode orders.

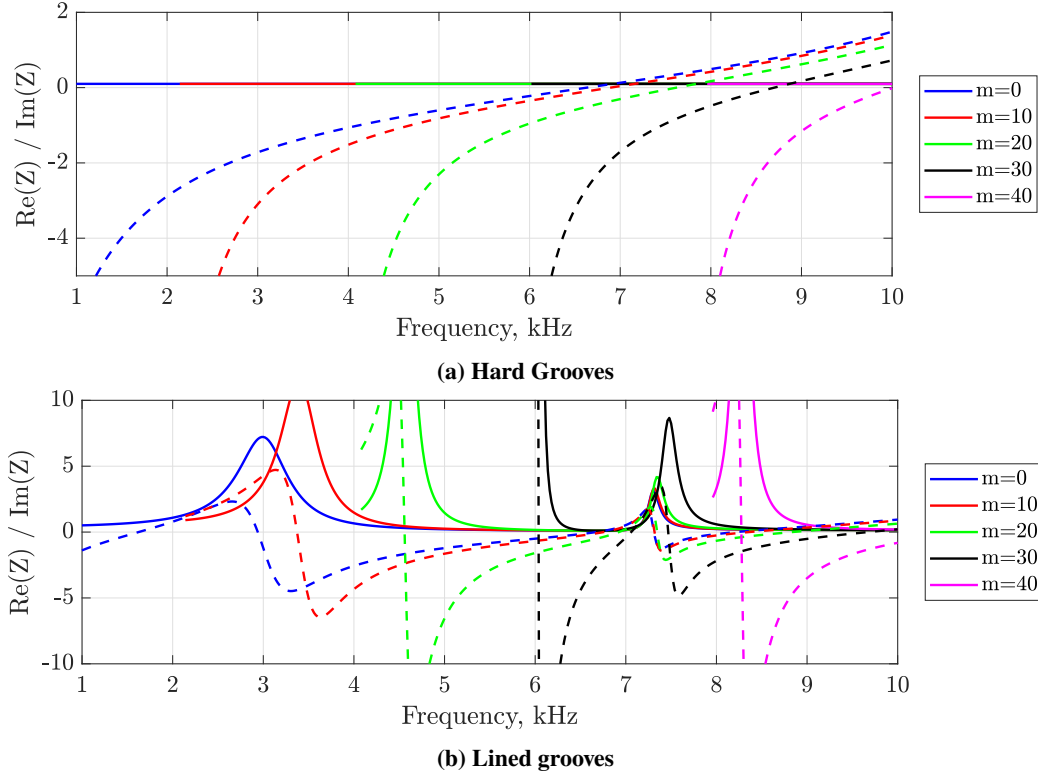


Fig. 14 Resistance and reactance of the OTR treatments predicted with our analytical groove impedance model for different azimuthal mode orders m : (a) Hard grooves and (b) Lined grooves. Results limited to the frequencies where the radial cavity modes are cut-on ($He > m$). Key: solid lines - resistance; dashed lines - reactance.

The measured SPL for the 'lined grooves', 'hard grooves' and 'hard wall' configurations are shown in Fig.15a. Background noise contaminating the data was reported in [8], in particular, two broadband humps that have been indicated in Fig.15a with blue shading. The hard grooves generate additional noise for frequencies over 4 kHz, which is attributed to the aerodynamic interaction of the fan tip and the grooves. The acoustic treatment in the lined grooves configuration attenuates part of the additional noise. However, the SPL is still higher than in the hard wall case for frequencies above 5 kHz. The area dominated by groove noise is indicated with red shading.

The SPL IL analytical estimates of the hard and lined grooves with respect to the hard wall baseline are compared with their experimental counterparts in Fig.15b. It can be observed from the comparison in the red shaded region that our linear acoustic model understandably cannot predict the noise amplification effects attributed to the fan interacting aerodynamically with the over-tip groove. This restricts the comparison of the analytical estimates and the experimental data to the frequency range of 1.5-3.5 kHz. In this range, the experimental data indicate that the hard groove have little impact on the SPL IL but the analytical estimates predict a noise amplification of ~ 1.5 dB. This may lead to the under-prediction of also ~ 1.5 dB that can be observed for the lined groove configuration. A subtraction of the hard groove and the lined groove SPL is performed with the aim of isolating the acoustic attenuation effects. That is, to eliminate any aerodynamic groove-related noise from the experimental data and leave only the effects of the acoustic treatment. This assumes that these effects are independent and therefore the impact of the grooves and the acoustic treatment can be addressed separately. The hard groove - lined groove results are presented for the non-contaminated frequency range of $1.5 \leq f \leq 3.5$ kHz in Fig.16.

It can be observed in Fig.16 that, apart from the BPF, a good agreement in the peak attenuation is obtained but with significant disagreement in the spectral shape. Please note that these estimates are obtained with a simple static point

source with zero mean flow. The results including flow and the inclusion of rotating incoherent sources will be reported in future publications. Further developments in the analytical modelling also include the implementation of a finite length liner and to include swirl in the fan section, which will aim at getting accurate and reliable OTR noise attenuation estimates.

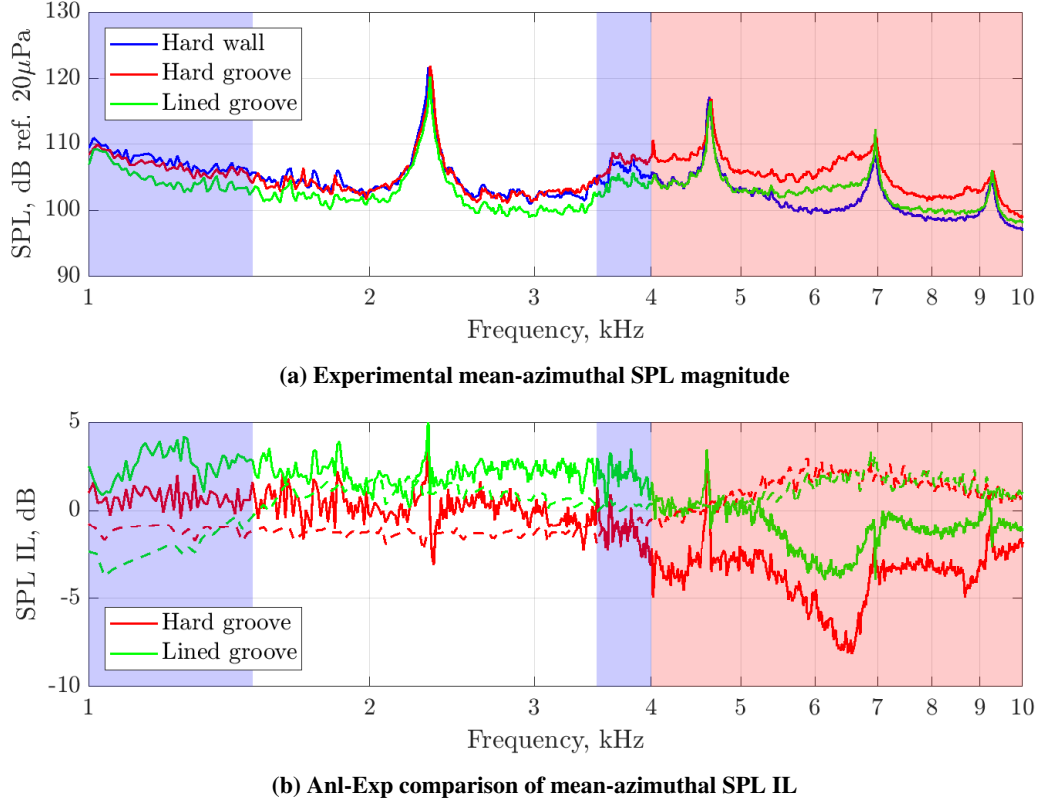


Fig. 15 (a) Experimental SPL magnitude (analysis bandwidth of 5 Hz) and (b) comparison of the experimental SPL IL with respect to the hard wall baseline and the analytical estimates for an axial dipole source located at $x_s/a = 0.068$ and $r_s/a = 0.95$ with $COR = 5$ and $M = 0$. Key Figure (b): solid lines - experimental; dashed lines - analytical. The blue shading indicates areas with background noise contamination and the red shading areas dominated by aerodynamic groove noise

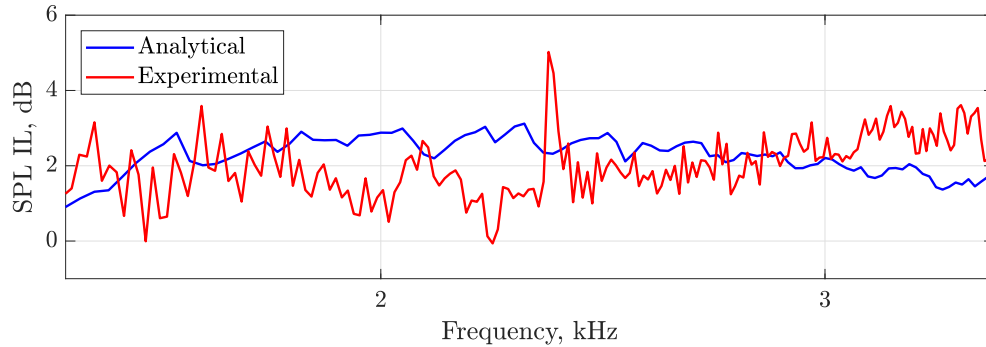


Fig. 16 Comparison of the hard groove - lined groove SPL IL obtained experimentally (analysis bandwidth of 5 Hz) and analytical estimates for an axial dipole source located at $x_s/a = 0.068$ and $r_s/a = 0.95$ with $COR = 5$ and $M = 0$.

VIII. Conclusions

This work has been focused on improving the understanding and prediction of the acoustic performance of Over-The-Rotor (OTR) liners, with the ultimate objective of using that understanding to develop a prediction method to guide the design of OTR liner designs for optimal noise reduction. The approach taken in this paper is analytical, based on the Green's function for an infinite lined cylindrical duct excited by monopole/dipole sources. Measured data published by NASA-GRC has been used to attempt a validation of our current analytical model.

The first analytical model, Model #1, comprised a monopole/dipole point source located in an infinite lined duct. Previous work showed that results agree with those published on the same model and the model has been cross-verified with FEM numerical solutions, both with and without mean flow. It has been shown that the power generated by a point monopole/dipole source in a duct depends on the impedance boundary condition, as well as the source excitation frequency and radial position, and this can lead to a negative liner insertion loss (IL) i.e. an amplification for some parameter combinations such as a liner impedance with a low resistance and positive reactance. Further work is required to understand this and to determine whether this could occur with real fan source distributions.

The second analytical model, Model #2, comprised a point source in an infinite duct with two sections, a semi-infinite hard wall and a semi-infinite lined wall section, the latter containing the point source. Predictions obtained with this model have shown that a point monopole source located at a realistic position relative to an OTR liner, of length equal to half the fan tip axial chord of a current large engine, could provide a peak broadband power insertion loss (IL) of around 4 dB, with significant attenuation over a wide range of source excitation frequencies. However, a more relevant point dipole source results in approximately 1 dB less attenuation than the monopole results. Direct comparison of Model #1 and Model #2 have indicated that the proximity of the source to a matching plane can cause back-reaction effects on the source that modify its power output. For a realistic position of the source relative to an OTR liner, these effects have been found to be significant (-0.6 to 1.4 dB).

The analytical Model #2 for OTR estimates coupled with the semi-locally reacting lined groove model has been compared with NASA OTR published experimental data. It is found that our acoustic model understandably cannot predict the noise amplification effects attributed to the fan interacting aerodynamically with the over-tip groove and hence causing a source modification. However, the estimates of the isolated acoustic attenuation of the treated grooves relative to the hard grooves present a good agreement in the peak attenuation but with significant disagreement in the spectral shape. Further work in increasing the complexity of the analytical models is required to obtain more accurate predictions of OTR liner acoustic performance.

Acknowledgments

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Appendix A: Mode Matching Procedure

The matching procedure is based on the continuity of mass and momentum at the matching interface plane proposed by Gabard and Astley [17]. The structure of this formulation is based on the comprehensive formulation of McAlpine et. al. [20] for a mode-matching technique based on continuity of pressure and axial particle velocity. A diagram of the interface and the nomenclature used are depicted in Fig.17.

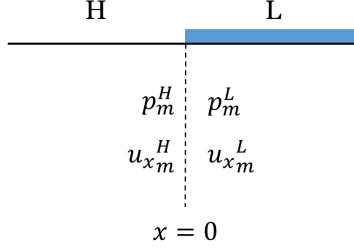


Fig. 17 Detail of the matching interface and nomenclature.

The impedance of the liner is assumed to be axisymmetric, which restricts any scattering between azimuthal modal orders. Under this condition, scattering is only possible between radial modes. Therefore, the matching problem can be solved separately for each azimuthal mode number m .

The time-harmonic pressure field in a generic cylindrical duct can be expressed as the superposition of right-running (+) and left-running (-) duct modes:

$$p(x, r, \theta) = \sum_{m=-\infty}^{\infty} e^{-jm\theta} \left(\sum_{n=1}^{\infty} A_{mn}^+ J_m(\alpha_{mn}^+ r) e^{-j\kappa_{mn}^+ x} + A_{mn}^- J_m(\alpha_{mn}^- r) e^{-j\kappa_{mn}^- x} \right) \quad (24)$$

The pressure field in the hard and lined ducts for each m is given by Eq.25-26. Note that the radial eigenvalues in the hard section are the same for both axial propagation directions ($\alpha_{mn}^{H+} = \alpha_{mn}^{H-} = \alpha_{mn}^H$).

$$p_m^H(x, r, \theta) = \sum_{n=1}^{\infty} \left(A_{mn}^{H+} J_m(\alpha_{mn}^H r) e^{-j\kappa_{mn}^{H+} x} + A_{mn}^{H-} J_m(\alpha_{mn}^H r) e^{-j\kappa_{mn}^{H-} x} \right) e^{-jm\theta} \quad (25)$$

$$p_m^L(x, r, \theta) = \sum_{n=1}^{\infty} \left(A_{mn}^{L+} J_m(\alpha_{mn}^L r) e^{-j\kappa_{mn}^{L+} x} + A_{mn}^{L-} J_m(\alpha_{mn}^L r) e^{-j\kappa_{mn}^{L-} x} \right) e^{-jm\theta} \quad (26)$$

The axial particle velocity can be obtained from the modal expansion of the pressure field and the non-dimensional linearised momentum equation in the axial direction:

$$j\omega u_x + M \frac{\partial u_x}{\partial x} + \frac{\partial p}{\partial x} = 0 \quad (27)$$

resulting for the hard and lined duct sections as

$$u_{x_m}^H(x, r, \theta) = \sum_{n=1}^{\infty} \left(\frac{\kappa_{mn}^{H+}}{\Omega_{mn}^{H+}} A_{mn}^{H+} J_m(\alpha_{mn}^H r) e^{-j\kappa_{mn}^{H+} x} + \frac{\kappa_{mn}^{H-}}{\Omega_{mn}^{H-}} A_{mn}^{H-} J_m(\alpha_{mn}^H r) e^{-j\kappa_{mn}^{H-} x} \right) e^{-jm\theta} \quad (28)$$

$$u_{x_m}^L(x, r, \theta) = \sum_{n=1}^{\infty} \left(\frac{\kappa_{mn}^{L+}}{\Omega_{mn}^{L+}} A_{mn}^{L+} J_m(\alpha_{mn}^L r) e^{-j\kappa_{mn}^{L+} x} + \frac{\kappa_{mn}^{L-}}{\Omega_{mn}^{L-}} A_{mn}^{L-} J_m(\alpha_{mn}^L r) e^{-j\kappa_{mn}^{L-} x} \right) e^{-jm\theta} \quad (29)$$

The continuity of mass and momentum across the matching interface is enforced by multiplying the mass and momentum equations by a continuous weighting function, then integrating them over a vanishingly small control volume. The detailed derivation can be consulted in reference [17].

The resultant non-dimensional corrected matching conditions with uniform mean flow are:

$$\int_S \bar{W}(p_2 - p_1) dS = -\frac{jM^2}{1-M^2} \frac{1}{\omega} \int_{\Gamma} \bar{W} \left(\frac{p_2}{Z_2} - \frac{p_1}{Z_1} \right) d\Gamma \quad (30)$$

$$\int_S \bar{W}(u_{x_2} - u_{x_1}) dS = \frac{jM}{1-M^2} \frac{1}{\omega} \int_{\Gamma} \bar{W} \left(\frac{u_{x_2}}{Z_2} - \frac{u_{x_1}}{Z_1} \right) d\Gamma \quad (31)$$

where S is the cross-sectional surface of the duct, Γ its contour, $\bar{W}$ a weighting function and the subscripts 1, 2 refer to each side of the matching plane. The terms on the right-hand side show a contribution of the impedance discontinuity in the presence of mean flow.

The matching equations for a hard-lined interface can be obtained by substituting the generic solutions of pressure and axial particle velocity into the expressions above. The hard wall eigenfunctions have been used as weighting functions in this instance. The matching equations for each azimuthal mode number m assuming a truncated value of radial modes N can be expressed as:

$$\int_{r=0}^1 J_m(\alpha_{mn}^H r) [p_m^L(0, r, \theta) - p_m^H(0, r, \theta)] r dr = -\frac{jM^2}{1-M^2} \frac{1}{\omega} J_m(\alpha_{mn}^H) \frac{p_m^L(0, 1, \theta)}{Z}, \quad n' = 1 \text{ to } N \quad (32)$$

$$\int_{r=0}^1 J_m(\alpha_{mn}^H r) [u_{x_m}^L(0, r, \theta) - u_{x_m}^H(0, r, \theta)] r dr = \frac{jM}{1-M^2} \frac{1}{\omega} J_m(\alpha_{mn}^H) \frac{p_m^L(0, 1, \theta)}{Z}, \quad n' = 1 \text{ to } N \quad (33)$$

The pressure and particle velocity at the hard and lined section evaluated at the matching plane are:

$$p_m^H(0, r, \theta) = \sum_{n=1}^N \left(A_{mn}^{H+} J_m(\alpha_{mn}^H r) e^{-j\kappa_{mn}^{H+} x_a} + A_{mn}^{H-} J_m(\alpha_{mn}^H r) \right) e^{-jm\theta} \quad (34)$$

$$p_m^L(0, r, \theta) = \sum_{n=1}^N \left(A_{mn}^{L+} J_m(\alpha_{mn}^{L+} r) + A_{mn}^{L-} J_m(\alpha_{mn}^{L-} r) e^{+j\kappa_{mn}^{L-} x_s} \right) e^{-jm\theta} \quad (35)$$

$$u_{x_m}^H(0, r, \theta) = \sum_{n=1}^N \left(\frac{\kappa_{mn}^{H+}}{\Omega_{mn}^{H+}} A_{mn}^{H+} J_m(\alpha_{mn}^H r) e^{-j\kappa_{mn}^{H+} x_a} + \frac{\kappa_{mn}^{H-}}{\Omega_{mn}^{H-}} A_{mn}^{H-} J_m(\alpha_{mn}^H r) \right) e^{-jm\theta} \quad (36)$$

$$u_{x_m}^L(0, r, \theta) = \sum_{n=1}^N \left(\frac{\kappa_{mn}^{L+}}{\Omega_{mn}^{L+}} A_{mn}^{L+} J_m(\alpha_{mn}^{L+} r) + \frac{\kappa_{mn}^{L-}}{\Omega_{mn}^{L-}} A_{mn}^{L-} J_m(\alpha_{mn}^{L-} r) e^{+j\kappa_{mn}^{L-} x_s} \right) e^{-jm\theta} \quad (37)$$

where x_s is the axial distance from the source to the matching plane and x_a the distance from the anechoic condition to the matching plane. The value assigned to x_a , in this case, is irrelevant because the anechoic condition implies that $A_{mn}^{H+} = 0$. The coefficients of the incoming modes from the source plane (A_{mn}^{L-}) can be obtained by direct comparison with Eq.1 for a monopole source or Eq.13 for a dipole source.

- Monopole point source:

$$A_{mn}^{L-} = -\frac{1}{2\pi j} \frac{J_m(\alpha_{mn}^{L-} r_s)}{Q_{mn}^{L-} J_m(\alpha_{mn}^{L-})^2} \quad (38)$$

- Dipole point source:

$$A_{mn}^{L-} = -\frac{1}{2\pi} \frac{J_m(\alpha_{mn}^{L-} r_s)}{Q_{mn}^{L-} J_m(\alpha_{mn}^{L-})^2} \left[\kappa_{mn}^{L-} \cos \gamma + \frac{m}{r_s} \sin \gamma \right] \quad (39)$$

with

$$Q_{mn}^{L-} = - \left[[\kappa_{mn}^{L-} + \Omega_{mn}^{L-} M] \left[1 - \left(\frac{m}{\alpha_{mn}^{L-}} \right)^2 - \frac{\Omega_{mn}^{\pm 4}}{(\omega \alpha_{mn}^{L-} Z)^2} \right] - \frac{2jM\Omega_{mn}^{L-}}{\omega Z} \right] \quad (40)$$

Substitution of Eq.34-37 into the matching equations leads to 2N sets of equations which can be used to obtain the 2N unknown coefficients, corresponding to the reflected (A_{mn}^{L+}) and transmitted (A_{mn}^{H-}) modes. The equations can be solved directly if arranged in the following form

$$\begin{bmatrix} \frac{A_{mn}^{H-}}{A_{mn}^{L+}} \end{bmatrix} = TD \begin{bmatrix} \frac{A_{mn}^{L-}}{A_{mn}^{H+}} \end{bmatrix} \quad (41)$$

T and D are 2N×2N transfer matrices. T is a square matrix which describes the pressure-velocity modal relations and the analytic integrals of the products of the Bessel functions. D is a diagonal matrix which contains the axial decay rates at the hard and lined duct sections. The matrices are given by:

$$T = \left[\begin{array}{c|c} a & -c^+ \\ \hline b^- & -d^+ \end{array} \right]^{-1} \left[\begin{array}{c|c} c^- & -a \\ \hline d^- & -b^+ \end{array} \right] \quad (42)$$

where each element (i, j) of the matrix is defined as

$$a_{ij} = \begin{cases} \int_{r=0}^1 J_m^2(\alpha_{mi}^H r) r dr & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (43)$$

$$b_{ij}^\pm = \begin{cases} \frac{\kappa_{mj}^{H\pm}}{\Omega_{mj}^{H\pm}} a_{ij} & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (44)$$

$$c_{ij}^\pm = \int_{r=0}^1 J_m(\alpha_{mi}^H r) J_m(\alpha_{mj}^{L\pm} r) r dr + M \varsigma_{ij}^\pm \quad (45)$$

$$d_{ij}^\pm = \frac{\kappa_{mj}^{H\pm}}{\Omega_{mj}^{H\pm}} \left[J_m(\alpha_{mi}^H r) J_m(\alpha_{mj}^{L\pm} r) r dr \right] - \varsigma_{ij}^\pm \quad (46)$$

and

$$D_{ii} = \begin{cases} e^{j\kappa_{mi}^{L-} x_s} & \text{if } i = 1, \dots, N \\ e^{-j\kappa_{mi}^{H+} x_a} & \text{if } i = N + 1, \dots, 2N \end{cases} \quad (47)$$

The term $\varsigma_{ij}^\pm$ represents the additional contributions one gets by using the continuity of mass and momentum at the matching interface instead of the standard condition of continuity of pressure and particle velocity. Therefore, this term is zero in the standard approach and is defined as follows when using continuity of mass and momentum:

$$\varsigma_{ij}^\pm = \frac{jM}{1-M^2} \frac{1}{\omega Z} J_m(\alpha_{mi}^H) J_m(\alpha_{mj}^{L\pm}) \quad (48)$$

The integrals of the products of the Bessel functions can be solved analytically by using

$$\int_0^a J_m^2(\alpha r_s) r_s dr_s = \frac{1}{2} J_m^2(\alpha a) \left(1 - \frac{m^2}{\alpha^2} \right) \quad (49)$$

$$\int_0^a J_m(\alpha r_s) J_m(\beta r_s) r_s dr_s = \frac{a}{\alpha^2 - \beta^2} \{ \beta J_m(\alpha a) J'_m(\beta a) - \alpha J'_m(\alpha a) J_m(\beta a) \} \quad (50)$$

Appendix B: Semi-locally reacting circumferential lined groove

This appendix describes the modelling of acoustically treated circumferential grooves as a semi-locally reacting liner. The acoustically treated circumferential grooves are depicted in Fig.18a, note that in the picture the cavity liner is filled with metal foam, the problem modelled here is equivalent but without the metal foam. Fig.18b shows a diagram of the problem and the nomenclature used in the derivation of the equivalent acoustic impedance. The two areas in Fig.18b are referred to (1) the 'treatment area', which consist of azimuthally partitioned cavity liners, and (2) the 'groove area', which does not present any azimuthal partition. Z_g is the total effective specific impedance of the whole liner (groove + cavity liner), Z_f is the specific impedance at the septum and a , d and h are the radius of the duct, groove depth and cavity depth respectively. In all analytical expressions of this section, the distances are normalised by a , the frequency by c_0/a , the pressure by $\rho_0 c_0^2$ and the particle velocity by c_0 . Therefore, the groove and cavity depth are non-dimensionalised with the duct radius a . The model is based on the following assumptions:

- Any mean axial or azimuthal flow in the groove is currently neglected. Further improvements of the model may consider flow in the azimuthal direction induced by the rotation of the fan blades over the treatment.
- The treatment area (1) behave as a SDOF cavity liner with impedance Z_f , any azimuthal and axial propagation is restricted by the cavity partitions and the grooves respectively.
- The grooves are non-locally reacting in the azimuthal direction.
- The azimuthal propagation in the groove must match that in the main duct. Therefore, the azimuthal mode number in the groove is the same than in the duct.
- The pressure field in the groove can be expressed as:

$$p(x, r, \theta) = \sum_{m=-\infty}^{\infty} e^{-jm\theta} \sum_{n=0}^{\infty} p_n(x) p_{mn}(r) \quad (51)$$

- The axial propagation in the groove is governed by a sum of cavity modes, where l_g is the non-dimensional groove-cavity width. Since the frequency required with the experimental geometry for the first axial mode to appear is $f \approx 30$ kHz, only the case of $n = 0$ has been considered, i.e the grooves are assumed to be axially locally reacting.

$$p_n(x) = A_n \cos(\kappa_n x) + B_n \sin(\kappa_n x), \quad \kappa_n = \frac{n\pi}{l_g} \quad (52)$$

- The radial propagation in the groove is governed by:
 - Pressure field of the form:

$$p_{mn}(r) = J_m(\alpha_n r) + K_{mn} Y_m(\alpha_n r) \quad (53)$$

where K_{mn} is an unknown constant dependent on m and n .

- The dispersion relation is reduced to $\alpha_n = \pm \sqrt{\omega^2 - \kappa_n^2} = \omega$, as it is assumed $n = 0$, where κ_n and α_n are the non-dimensional axial and radial wave numbers.
- The boundary condition at the septum is:

$$\frac{p|_{r=1+d}}{u_r|_{r=1+d}} = Z_f \quad (54)$$

The radial particle velocity can be obtained from Eq. 53:

$$u_{r_n}(r) = \frac{j}{\omega} \frac{\partial p}{\partial r} = \frac{j\alpha_n}{\omega} [J'_m(\alpha_n r) + K_{mn} Y'_m(\alpha_n r)] \quad (55)$$

By using the dispersion relation ($\alpha = \omega$), K_{mn} is now only dependent on m and can be determined by applying the boundary condition at the septum, that is, substituting Eq. 53 and Eq. 55 into Eq. 54.

$$K_m = - \frac{J'_m(\omega[1+d]) + j \frac{J_m(\omega[1+d])}{Z_f}}{Y'_m(\omega[1+d]) + j \frac{Y_m(\omega[1+d])}{Z_f}} \quad (56)$$

Finally, the total effective impedance of the treatment results in

$$Z_g(\omega, m) = \frac{p|_{r=1}}{u_r|_{r=1}} = -j \frac{J_m(\omega) + K_m Y_m(\omega)}{J'_m(\omega) + K_m Y'_m(\omega)} \quad (57)$$

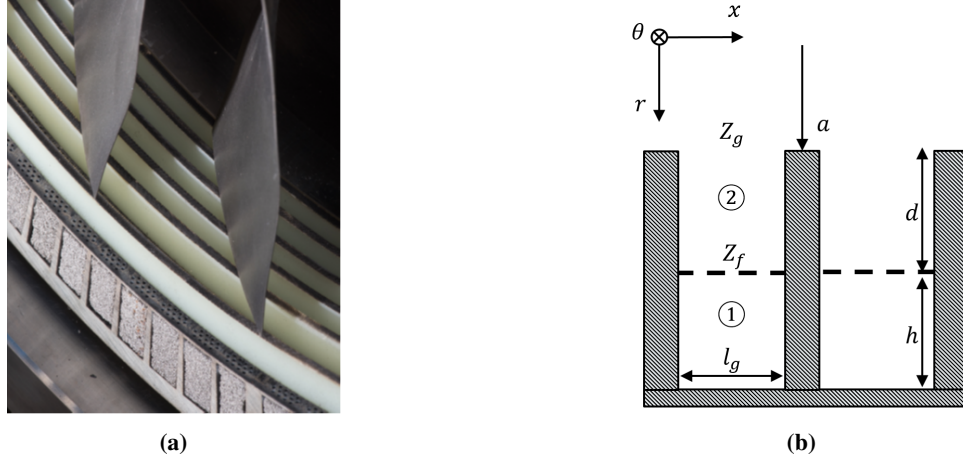


Fig. 18 OTR fan case liner: (a) view of the liner installed in the NASA W-8 fan rig [8] and (b) diagram of the problem and nomenclature.

Since the grooves are assumed to be locally reacting in the axial direction but non-locally reacting in the azimuthal direction, the impedance model has been tentatively called as *semi-locally reacting*. The configuration of grooves without treatment, i.e. hard wall bottom, can be modelled by changing the septum boundary condition to $\left. \frac{\partial p}{\partial r} \right|_{r=1+d} = 0$ and repeat the process or simply by setting $Z_f = \infty$ in Eq.57.

An alternative approximated expression can be obtained by assuming that the radius of the duct is much bigger than the groove depth, i.e $d \ll 1$. Under that assumption, the annular casing can be approximated by a linear section as indicated in Fig.19. Neglecting again any axial propagation, the pressure field in the groove can be expressed by:

$$p(r, \theta) = p(r)e^{jm\theta} \approx p(y, z) = p(y)e^{jk_z z} \quad (58)$$

with the non-dimensional wave number in the z-direction k_z as

$$k_z = m \quad (59)$$

The solution of the Helmholtz equation in the grooves is of the form

$$p_m(y) = A_m \cos(k_y y) + B_m \sin(k_y y) \quad (60)$$

where the non-dimensional wave number in the y-direction k_y can be obtained from the dispersion relation

$$k_y^2 = \omega^2 - k_z^2 = \omega^2 - m^2 \quad (61)$$

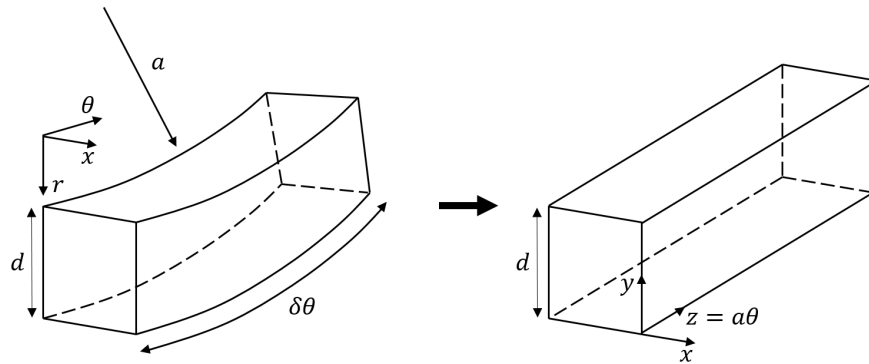


Fig. 19 Approximation of the annular grooves into a linear section when $d \ll 1$.

Following an analogous procedure than in the previous approach with the annular groove, the expression for the effective impedance of the treatment (Eq. 62) can be obtained by applying the boundary condition at the bottom of the groove ($y = 0$) and evaluating the pressure and particle velocity in the interface with the main duct ($y = d$). Note that the y -axis is defined in the opposite direction of the radius.

$$Z_g(\omega, m) = -\frac{j\omega}{k_y} \left[\frac{\cot(k_y d) + \frac{j\omega}{Z_f k_y}}{1 - \frac{j\omega}{Z_f k_y} \cot(k_y d)} \right] \quad (62)$$

Once again, the configuration of hard grooves can be modelled by changing the septum boundary condition to $\frac{\partial p}{\partial r}\big|_{y=0} = 0$ and repeat the process or simply by setting $Z_f = \infty$ in Eq.62. As it can be observed in the expressions of the equivalent impedance, Eq.57 and Eq.62, the acoustic response of the grooves (hard or lined) is not only a function of frequency but also of the azimuthal mode number m .

To assess the accuracy of the approximation of the annular grooves as a linear section for this particular application, Fig.20 compares the impedance obtained with each model for the range of frequencies used in the experimental data ($0 \leq He \leq 50$) and for the values of interest of the liner parameters shown in Table 2. The impedance at the septum (Z_f) is obtained using Eq.19 and the rig duct radius is $a = 279.4$ mm, giving a non-dimensional groove depth of ~ 0.046 .

The agreement shown in Fig.20 for $m = 22$ is consistent with the results at lower and higher azimuthal mode numbers. The two impedance models present a good agreement for the whole range of frequencies used in this research for both a hard (Fig.20a) and a lined (Fig.20b) groove. Model A has therefore served the useful purpose of checking the validity of Model B and in the current work the former is used as the default method.

Table 3 Numerical values used for the impedance parameters.

R_{fs}	t_{fs}	d_{fs}	σ	h	d
0.5	1.5 mm	0.89 mm	0.1	25.4 mm	12.7 mm

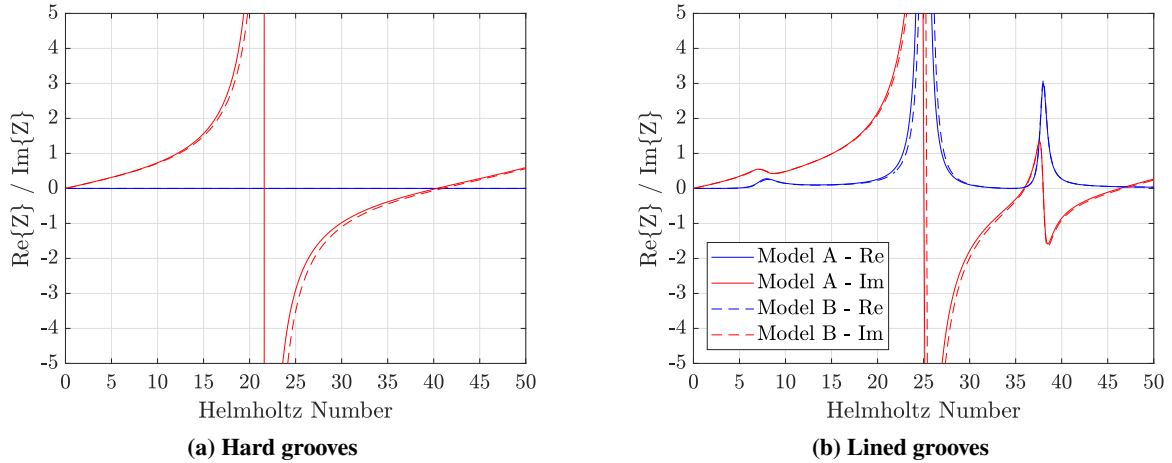


Fig. 20 Comparison of the impedance models based on the annular (Model A) and linear groove (Model B).

References

- [1] Sutliff, D. L., and Jones, M. G., “Low-Speed Fan Noise Attenuation from a Foam-Metal Liner,” *Journal of Aircraft*, Vol. 46, No. 4, 2009, pp. 1381–1394. doi:10.2514/1.41369, URL <http://arc.aiaa.org/doi/10.2514/1.41369>.
- [2] Elliott, D., Woodward, R., and Podboy, G., “Acoustic Performance of Novel Fan Noise Reduction Technologies for a High Bypass Model Turbofan at Simulated Flight Conditions,” *15th AIAA/CEAS Aeroacoustics Conference (30th AIAA Aeroacoustics Conference)*, , No. May, 2009, pp. 11–13. doi:10.2514/6.2009-3140, URL <http://arc.aiaa.org/doi/abs/10.2514/6.2009-3140>.
- [3] Sutliff, D. L., Elliott, D. M., Jones, M. G., and Hartley, T. C., “Attenuation of FJ44 turbofan engine noise with a foam-metal liner installed over-the-rotor,” *AIAA Paper*, Vol. 3141, No. May, 2009, p. 2009. doi:10.2514/6.2009-3141.
- [4] Sutliff, D. L., Jones, M. G., and Hartley, T. C., “High-Speed Turbofan Noise Reduction Using Foam-Metal Liner Over-the-Rotor,” *Journal of Aircraft*, Vol. 50, No. 5, 2013, pp. 1491–1503. doi:10.2514/1.C032021, URL <http://arc.aiaa.org/doi/10.2514/1.C032021>.
- [5] Hughes, C. E., and Gazzaniga, J. A., “Effect of Two Advanced Noise Reduction Technologies on the Aerodynamic Performance of an Ultra High Bypass Ratio Fan,” *15th AIAA/CEAS Aeroacoustics Conference (30th AIAA Aeroacoustics Conference)*, , No. May, 2009, pp. 11–13. doi:10.2514/6.2009-3139, URL <http://dx.doi.org/10.2514/6.2009-3139>.
- [6] Bozak, R., Hughes, C., and Buckley, J., “GT2013-95114,” *The Aerodynamic Performance of an Over-The-Rotor Liner with Circumferential Grooves on a High Bypass Ratio Turbofan Rotor*, 2013, pp. 1–8.
- [7] Gazella, M. R., Takakura, T., Sutliff, D. L., Bozak, R., and Tester, B. J., “Evaluating the Acoustic Benefits of Over-the-Rotor Acoustic Treatments Installed on the Advanced Noise Control Fan,” *AIAA/CEAS Aeroacoustics Conference*, , No. 23RD, 2017.
- [8] Bozak, R., and Dougherty, R. P., “Measurement of Noise Reduction from Acoustic Casing Treatments Installed Over a Subscale High Bypass Ratio Turbofan Rotor,” *2018 AIAA/CEAS Aeroacoustics Conference*, 2018, pp. 1–14. doi:10.2514/6.2018-4099, URL <https://arc.aiaa.org/doi/10.2514/6.2018-4099>.
- [9] Jones, M. G., and Howerton, B. M., “Evaluation of Novel Liner Concepts for Fan and Airframe Noise Reduction,” *22nd AIAA/CEAS Aeroacoustics Conference*, 2016, pp. 1–18. doi:10.2514/6.2016-2787, URL <http://arc.aiaa.org/doi/10.2514/6.2016-2787>.
- [10] Rienstra, S. W., and Tester, B. J., “An analytic Green’s function for a lined circular duct containing uniform mean flow,” *Journal of Sound and Vibration*, Vol. 317, No. 3-5, 2008, pp. 994–1016. doi:10.1016/j.jsv.2008.03.048.
- [11] Rienstra, S. W., “A classification of duct modes based on surface waves,” *Wave Motion*, Vol. 37, No. 2, 2003, pp. 119–135. doi:10.1016/S0165-2125(02)00052-5.
- [12] Lowis, C. R., and Joseph, P. F., “Determining the strength of rotating broadband sources in ducts by inverse methods,” *Journal of Sound and Vibration*, Vol. 295, No. 3-5, 2006, pp. 614–632. doi:10.1016/j.jsv.2006.01.031.
- [13] Morfey, C. L., “Sound transmission and generation in ducts with flow,” *Journal of Sound and Vibration*, Vol. 14, No. 1, 1971, pp. 37–55. doi:10.1016/0022-460X(71)90506-2.
- [14] Morfey, C. L., “Acoustic energy in non-uniform flows,” *Journal of Sound and Vibration*, Vol. 14, No. 2, 1971, pp. 159–170. doi:10.1016/0022-460X(71)90381-6.
- [15] Eversman, W., “Energy flow criteria for acoustic propagation in ducts with flow,” *The Journal of the Acoustical Society of America*, Vol. 49, 1971, p. 1717. doi:10.1121/1.1912567.
- [16] Guerin, S., Moreau, A., and Tapken, U., “Relation between source models and acoustic duct modes,” *AIAA Conference*, , No. May, 2009, p. 3364. doi:10.2514/6.2009-3364.
- [17] Gabard, G., and Astley, R. J., “A computational mode-matching approach for sound propagation in three-dimensional ducts with flow,” *Journal of Sound and Vibration*, Vol. 315, No. 4-5, 2008, pp. 1103–1124. doi:10.1016/j.jsv.2008.02.015.
- [18] Lansing, D. L., and Zorumski, W. E., “Effects of wall admittance changes on duct transmission and radiation of sound,” *Journal of Sound and Vibration*, Vol. 27, No. 1, 1973, pp. 85–100.
- [19] Unruh, J. F., “Finite length tuning for low frequency lining design,” *Journal of Sound and Vibration*, Vol. 45, No. 1, 1976, pp. 5–14.
- [20] McAlpine, A., Astley, R. J., Hii, V. J., Baker, N. J., and Kempton, A. J., “Acoustic scattering by an axially-segmented turbofan inlet duct liner at supersonic fan speeds,” *Journal of Sound and Vibration*, Vol. 294, No. 4, 2006, pp. 780–806. doi:10.1016/j.jsv.2005.12.039.